

Arithmetic Operators – Real-Life Problem Statements

1. Monthly Salary Calculation

Scenario:

A company pays an employee a fixed monthly salary and additional incentives based on performance.

Problem Statement:

Write a Python program to calculate the total monthly salary of an employee by adding the fixed salary and incentive amount.

Input:

- Basic Salary
- Incentive

Output:

- Total Salary
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2. Grocery Store Bill

Scenario:

A customer purchases multiple packets of rice from a grocery store.

Problem Statement:

Write a Python program to calculate the total cost of rice packets purchased.

Input:

- Price per packet
- Number of packets

Output:

- Total Bill Amount
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3. Fuel Consumption Tracker

Scenario:

A person wants to calculate the average fuel consumption of a car.

Problem Statement:

Write a Python program to find the average mileage of a car.

Input:

- Total distance traveled (km)
- Fuel consumed (liters)

Output:

- Mileage (km/l)
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4. Bank Account Balance**Scenario:**

A customer withdraws money from their bank account.

Problem Statement:

Write a Python program to calculate the remaining balance after withdrawal.

Input:

- Current Balance
- Withdrawal Amount

Output:

- Remaining Balance
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5. Classroom Seating Arrangement**Scenario:**

A school wants to arrange students into equal rows.

Problem Statement:

Write a Python program to determine how many complete rows can be formed.

Input:

- Total Students
- Students per Row

Output:

- Number of Complete Rows
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6. Pizza Distribution**Scenario:**

A party organizer wants to distribute pizza slices equally among children.

Problem Statement:

Write a Python program to find how many slices remain after equal distribution.

Input:

- Total Pizza Slices

- Number of Children

Output:

- Remaining Slices
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7. Compound Growth of Savings

Scenario:

A person invests money and wants to see how the amount grows over years.

Problem Statement:

Write a Python program to calculate the value of money after a certain number of years assuming it doubles every year.

Input:

- Initial Amount
- Number of Years

Output:

- Final Amount
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8. Online Shopping Discount

Scenario:

An e-commerce website offers a fixed discount on products.

Problem Statement:

Write a Python program to calculate the final payable amount after applying the discount.

Input:

- Product Price
- Discount Amount

Output:

- Final Price
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9. Mobile Recharge Plan

Scenario:

A telecom company charges a fixed amount per GB of data.

Problem Statement:

Write a Python program to calculate the total recharge amount based on the data pack selected.

Input:

- Cost per GB
- Number of GBs

Output:

- Total Recharge Cost
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10. Event Management Budget**Scenario:**

An event manager wants to calculate the cost contribution per participant.

Problem Statement:

Write a Python program to calculate how much each participant should pay.

Input:

- Total Event Cost
- Number of Participants

Output:

- Amount per Participant
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Food Delivery Business Profit Analysis

Scenario:

A food delivery startup wants to calculate its daily profit and package distribution details.

Problem Statement:

Write a Python program that:

1. Calculates total revenue generated.
2. Calculates profit after deducting expenses.
3. Determines how many complete delivery boxes can be prepared.
4. Finds remaining food packets.
5. Predicts revenue after a certain number of days assuming revenue doubles daily.

Input:

- Number of orders
- Price per order
- Daily expenses
- Total food packets
- Packets per box
- Number of days

Output:

- Total Revenue
- Profit
- Complete Boxes
- Remaining Packets
- Future Revenue